ANALYTICAL CUMULATIVE EXAM
February 25, 2000

1. (40 points) State whether the following factors will cause a positve, negative or insignificant deviation from Beer's Law. Explain your answers. Assume that the measurements are not in the ultratrace region.

   a. Stray light reaching the photodetector
   b. Fluorescence
   c. Raman Scattering
   d. A change in pH from 7 to 2 while operating at the isobestic point of an organic acid
   e. Nonmonochromatic radiation

2. (40 points) Given the data to the right, estimate the signal to noise ratio for the least concentrated standard, and the sensitivity and detection limit (in units of molarity) for the technique. State your specifications for each calculation. The vertical scale is in units of absorbance. This data was taken with a one second time constant (sampling interval). What would the detection limit be with a 10 second time constant?

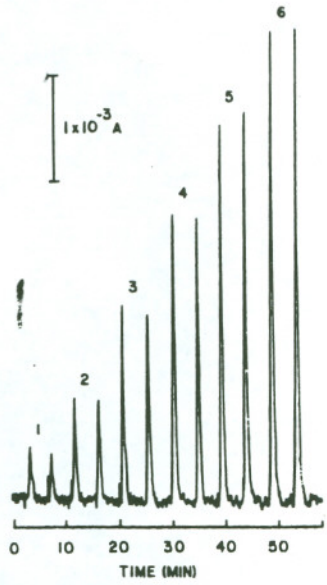

Figure 2. Flow injection peaks of bromophenol blue/ethanol in duplicate at (1) 1.0×10^{-6} M, (2) 2.0×10^{-6} M, (3) 4.0×10^{-6} M, (4) 6.0×10^{-6} M, (5) 8.0×10^{-6} M, and (6) 1.0×10^{-5} M.

3. (20 points) Given the following information, what is the smallest power, in watts, that can be detected by this light detection system.

   Planck's constant: 6.63×10^{-34} Js
   Speed of Light: 3×10^{10} cm/s
   Charge on an electron: 1.6×10^{-19} coulombs
   Time constant: one second (sampling time)
   Wavelength of Measurement: 500 nm
   Smallest measureable current: one nanoamp (10^{-9} A)
   Photomultiplier tube specifications:
   - type: Thorn #9816
   - operating voltage: 1800 volts, gain @ 1.8 KV is 1×10^{6}
   - photocathode: S-20 type, quantum efficiency @ 500 nm is 0.15